SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Data Warehousing and Data Mining
COURSE CODE: CSA1603

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction , for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding *(2)*	Method Selection *(2)*	Solution Process *(2.5)*	Accuracy of Calculation *(2)*	Interpretation of Results *(1.5)*	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Nissanki Sri Venkata Lakshmi Kavya (192373036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
N.S.V.L.Kavya

Annexure B

Analytical Problem Solving

1. Covariance Analysis in Health Insurance

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.

Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	Claims per 100 Policyholders (yi)
3.2	15
4.0	18
5.5	25
4.8	20

Conclusion: Covariance is positive (3.0625), indicating a positive relationship between premium increase and claim frequency in this dataset.

Sum of deviation products = 12.25. Cov(x,y) = 12.25 / 4 = 3.0625.

Deviation products: (3.2−4.375)(15−19.5)=5.2875; (4.0−4.375)(18−19.5)=0.5625; (5.5−4.375)(25−19.5)=6.1875; (4.8−4.375)(20−19.5)=0.2125.

Covariance formula (population): $Cov(x,y) = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / n$.

Mean of y: $\bar{y} = (15 + 18 + 25 + 20) / 4 = 78 / 4 = 19.5$.

Mean of x: $\bar{x} = (3.2 + 4.0 + 5.5 + 4.8) / 4 = 17.5 / 4 = 4.375$.

Given x = 3.2, 4.0, 5.5, 4.8 and y = 15, 18, 25, 20.

2. Data Smoothing using Equal-Frequency Binning (Healthcare)

Sample Data Set (Systolic BP readings):

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

Interpretation: Equal-frequency binning groups the sorted observations into equal-sized bins and reduces minor fluctuations.

Alternative smoothing by medians: replace all Bin 1 values with 118, Bin 2 values with 126, and Bin 3 values with 145.

Smoothing by bin means: replace all Bin 1 values with 117.33, Bin 2 values with 129.56, and Bin 3 values with 149.00.

Bin 3: 138, 138, 138, 140, 145, 150, 152, 160, 180. Mean = 149.00; median = 145.

Bin 2: 124, 126, 126, 126, 126, 130, 135, 135, 138. Mean = 129.56; median = 126.

Bin 1: 110, 112, 115, 115, 118, 120, 120, 122, 124. Mean = 117.33; median = 118.

Assumption: The question does not specify the number of bins. Using 3 equal-frequency bins, 27 readings give 9 readings per bin.

3. Equal-Width Binning for Risk Categorization

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.

To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Interpretation: Equal-width binning divides the BMI range into intervals of equal numerical width; each value is replaced with its nearest boundary.
Smoothed sequence: 18.5, 18.5, 18.5, 25.5, 25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 39.5, 39.5.
Bin 3: 33.1, 33.6, 34.7, 38.2, 39.5 → 32.5, 32.5, 32.5, 39.5, 39.5.
Bin 2: 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4 → 25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5.
Bin 1: 18.5, 19.2, 21.8, 24.9 → 18.5, 18.5, 18.5, 25.5.
Smoothing to the nearest boundary:
Using 3 bins, bin width = 21 / 3 = 7. Intervals: Bin 1 = 18.5–25.5; Bin 2 = 25.5–32.5; Bin 3 = 32.5–39.5.
Minimum BMI = 18.5; maximum BMI = 39.5; range = 39.5 − 18.5 = 21.

4. Data Reduction using Bin Medians (Medical Supply Costs)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**.
To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.
Sample Data Set (Costs in ₹):
120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

5. Standardization of Patient Age Data

Given the following **ages of insured individuals**:
18, 22, 25, 42, 28, 43, 33, 35, 56, 28
Standardize the variable by performing the following:
(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.
(a) Mean absolute deviation (MAD)
Ages: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28.
Mean = (18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28) / 10 = 330 / 10 = 33 years.
Absolute deviations from the mean: 15, 11, 8, 9, 5, 10, 0, 2, 23, 5. Sum = 88.
MAD = 88 / 10 = 8.8 years.
(b) Z-scores for the first four observations
Using population standard deviation: $\sigma = \sqrt{[\Sigma(x_i - \text{mean})^2 / n]} = \sqrt{117.4} \approx 10.8351$.
Formula: $z = (x - \text{mean}) / \sigma$, with mean = 33.
Age 18: $z = (18 - 33) / 10.8351 \approx -1.38$.
Age 22: $z = (22 - 33) / 10.8351 \approx -1.02$.
Age 25: $z = (25 - 33) / 10.8351 \approx -0.74$.
Age 42: $z = (42 - 33) / 10.8351 \approx 0.83$.

Final answer: MAD = 8.8 years; first four z-scores are −1.38, −1.02, −0.74, and 0.83.

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation